Photosynthesis: An Overview

Photosynthesis is the process by which green plants, algae and some bacteria convert light energy into chemical energy stored in glucose. It takes place inside the chloroplasts of plant cells, where the pigment chlorophyll absorbs light most strongly in the blue and red regions of the visible spectrum.

The overall balanced equation is:
6 CO2 + 6 H2O + light -> C6H12O6 + 6 O2

The Light-Dependent Reactions

The light-dependent reactions occur in the thylakoid membrane of the chloroplast. Photosystem II splits water molecules, releasing oxygen as a by-product and supplying electrons to the transport chain.

As electrons move down the chain, proton gradients drive ATP synthase to produce ATP. Photosystem I reduces NADP+ to NADPH. The ATP and NADPH produced here are consumed by the Calvin cycle.

The Calvin Cycle

The Calvin cycle takes place in the stroma of the chloroplast and does not require light directly. It fixes carbon dioxide into an organic molecule using the enzyme RuBisCO, the most abundant protein on Earth.

Each turn of the cycle consumes three ATP and two NADPH, and three turns are required to produce one molecule of glyceraldehyde-3-phosphate and ultimately glucose.

Limitations and Open Questions

The efficiency of photosynthesis is limited by several factors: light intensity, carbon dioxide concentration and temperature. Photorespiration in C3 plants can waste up to a quarter of the fixed carbon.

This review does not address C4 and CAM adaptations, which reduce photorespiration in hot, dry environments. Quantitative modelling of the light reactions is left for future work.